

Usecase name	Login/Register
Actor	Passenger
preconditions	• The user has access to the system. • The system is running.
Flow of events	1. The user navigates to the login/register page. 2. For registration: • The user selects “Register.” • The user enters required information (username, email, password). • The system validates the information. • The system creates a new account. 3. For login: • The user enters username/email and password. • The system verifies the credentials.
Postconditions	• The user is authenticated and logged in. • The user can access system features.
exeption	• Invalid registration data (e.g., email already exists): System displays an error. • Invalid login credentials: System displays an error.

Usecase name	View Train Information
Actor	Passenger
preconditions	• The user is logged in or has access to the system. • The train database is up to date.
Flow of events	1. The user selects “View Train Information.” 2. The system retrieves train data. 3. The system displays train schedules, routes, and details. 4. The user can filter or search for specific trains.
Postconditions	he user views the available train information.
exeption	• No trains match the search/filter: System shows a message. • System/database error: System displays an error message.

Usecase name	Check Seat Availability
Actor	Passenger
preconditions	• The user is logged in. • The user has selected a train or route.
Flow of events	1. The user selects “Check Availability” for a specific train. 2. The system queries the database for available seats. 3. The system displays available seats, classes, and prices.
Postconditions	• The user sees the seat availability for the selected train.
exeption	• No seats available: System notifies the user and may suggest alternatives. • System/database error: System displays an error message.

Usecase name	Book Ticket
Actor	Passenger
preconditions	• The user is logged in. • The user has selected a train with available seats.
Flow of events	1. The user selects “Book Ticket.” 2. The user chooses seat preferences (class, number of seats). 3. The system confirms real-time availability. 4. The user enters passenger details. 5. The system generates a booking summary with total fare. 6. The user proceeds to payment. 7. The system processes payment. 8. The system issues the ticket.
Postconditions	• The ticket is booked and issued. • The selected seats are marked as reserved.
exeption	• Seats become unavailable during booking: System notifies the user. • Payment fails: System notifies the user and prompts for retry.

Usecase name	Add Additional Services
Actor	Passenger
preconditions	- The user has a confirmed booking or is in the booking process. - Additional services are available for the selected train.
Flow of events	- The user selects “Add Additional Services.” - The system displays available options (meals, extra luggage, etc.). - The user selects desired services. - The system updates the total fare. - The user confirms the selection.
Postconditions	- Additional services are added to the booking. - The fare is updated accordingly.
exeption	Selected service is unavailable: System notifies the user and prompts for alternatives.

Usecase name	Cancel Ticket
Actor	Passenger
preconditions	- The user is logged in. - The user has an active booking.
Flow of events	- The user selects “Cancel Ticket.” - The system shows active bookings. - The user selects a booking to cancel. - The system displays cancellation policy and refund amount. - The user confirms cancellation. - The system processes the cancellation.
Postconditions	- The ticket is cancelled. - The seat is released. - Refund process is initiated (if applicable).
exeption	- Booking is not eligible for cancellation: System notifies the user. - Cancellation period has expired: System notifies the user.

Usecase name	Request Healthcare/Meals
Actor	Passenger
preconditions	- The user has a confirmed booking. - Healthcare/meals options are available for the selected train.
Flow of events	- The user selects “Request Healthcare/Meals.” - The system displays available options. - The user makes selections. - The system updates the booking.
Postconditions	Preferences are recorded in the booking.
exeption	Selected options are unavailable: System notifies the user.

Usecase name	Edit Personal Details
Actor	admin
preconditions	The user is logged in.
Flow of events	- The user navigates to “Personal Details.” - The system displays current information. - The user updates details. - The system validates and saves changes.
Postconditions	Personal details are updated.
exeption	Invalid information: System prompts user to correct errors.

Usecase name	Verify Travel Details
Actor	Verification manager
preconditions	- The user is in the booking process. - Travel details have been entered.
Flow of events	- The system displays entered travel details. - The user reviews and confirms or edits details. - The system validates the details.
Postconditions	Travel details are confirmed and correct.

exeption	Invalid or incomplete details: System prompts user to correct.
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Usecase name	Ticket Fare and Payment
Actor	Payment method
preconditions	The user has selected seats and any additional services.
Flow of events	- The system calculates and displays the total fare. - The user reviews fare breakdown. - The user selects a payment method. - The system processes the payment.
Postconditions	- Payment is successful. - Booking is confirmed and ticket is issued.
exeption	Payment fails: System notifies user and allows retry.

Usecase name	Check Booked Tickets
Actor	Verification manager
preconditions	The user is logged in.
Flow of events	- The user selects “Check Booked Tickets.” - The system retrieves and displays the user's bookings. - The user can view details of each booking.
Postconditions	The user can view booking history and details
exeption	No bookings found: System notifies the user.

Usecase name	Apply for Refund
Actor	passenger
preconditions	- The user has cancelled a booking. - The booking is eligible for a refund.
Flow of events	- The user selects “Apply for Refund.” - The system displays eligible cancelled bookings. - The user selects a booking. - The system shows refund amount and policy. - The user confirms the request. - The system processes the refund request.

Postconditions	Refund request is submitted and tracked.
exeption	- Booking not eligible for refund: System notifies the user. - Refund period expired: System notifies the user.

Usecase name	Check Refund Status
Actor	Verification manager
preconditions	The user has submitted a refund request.
Flow of events	- The user selects “Check Refund Status.” - The system displays all refund requests. - The user selects a request to view its status.
Postconditions	The user can see the current status of their refund.
exeption	No refund requests found: System notifies the user.

Usecase name	Select Payment Method
Actor	-
preconditions	The user is ready to pay for a booking.
Flow of events	- The system displays available payment methods. - The user selects a method and enters required details. - The system validates and confirms the payment method.
Postconditions	Payment method is selected and ready for payment processing.
exeption	Selected payment method is unavailable: System notifies the user to choose another.